FSOC Data Rate calculations.

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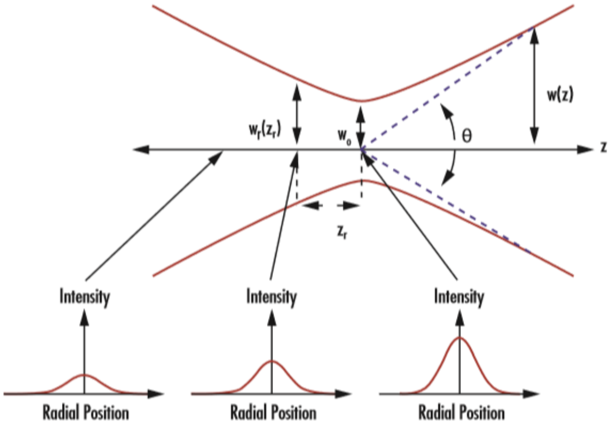
Laser beam propagation[1]



In the plot, $W_r(z_r)$ belongs to beam width at Rayleigh distance z_r, W_0 to the beam waist and $W(z)$ to the beam width at any distance z, and θ the beam divergence.

Divergence

$$\theta = \frac{\lambda}{\pi W_0}$$

Spot size

With $W_0 = w_0$, the beam spot size is given by:

$$w(z) \approx w_0 + z\theta$$

with differs from beam diameter, which is $2w(z)$ using this convention referred to the graphic.

Field of View (FOV)

$$\mathrm{FOV} = 2\alpha = 2\tan^{-1}\left(\frac{d}{2f}\right) \approx \frac{d}{f}$$

Gaussian beam

$$U_0(r, 0) = \exp\left(-\frac{r^2}{W_0^2} - \frac{ikr^2}{2F_0}\right)$$

$$U_0(r, L) = \frac{1}{p(L)} \exp(ikL) \exp\left(-\frac{r^2}{W_0^2} - \frac{ikr^2}{2F_0}\right)$$

$$p(L) = 1 - \frac{L}{F_0} + i\frac{2L}{kW_0^2}$$

[1] most equations according to Andrew-Phillips book.

Gaussian beam parameters

$$\Theta_0 = 1 - \frac{L}{F_0}$$

$$\Lambda_0 = \frac{2L}{kW_0^2}$$

Gaussian Beam width

$$W = W_0\sqrt{\Theta_0^2 + \Lambda_0^2}$$

Irradiance on-axis

$$I(0, L) = I_0 \cdot \frac{W_0^2}{W^2} \left[\frac{\mathrm{W}}{\mathrm{m}^2}\right]$$

Power at the receiver

$$P_{R_x} = P_{T_x} \cdot \left(1 - e^{\frac{-2r^2}{W^2}}\right) [\mathrm{W}]$$

Irradiance at the receiver

$$I_{R_x} = 4\frac{P_{R_x}}{\pi D_{R_x}^2} \left[\frac{\mathrm{W}}{\mathrm{m}^2}\right]$$

Atmospheric channel effects

Kolmogorov outer scale and inner scale

Inertial range:

$$D_{RR}(R) = \left\langle (V_1 - V_2)^2 \right\rangle = C_V^2 R^{2/3}$$

$$D_{RR}(R) = \begin{cases} C_v^2 \ell_0^{-4/3} R^2 & , \ 0 < R << \ell_0 \\ C_v^2 R^{2/3} & , \ \ell_0 << R << L_0 \end{cases}$$

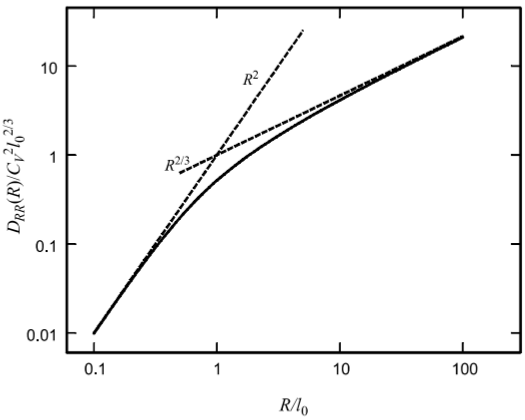


Figure 3.2 The scaled longitudinal velocity structure function showing both R^2 and $R^{2/3}$ asymptotic domains (dashed lines).

Beam spreading (Spot size)

Long-term spot size W_{LT}:

$$W_{\text{LT}}^2 = W^2 + W^2 T_{SS} + \langle r_c^2 \rangle = W_{\text{ST}}^2 + \langle r_c^2 \rangle$$

with $\langle r_c^2 \rangle$ the beam wander displacement variance.

Loss (Uplink):

$$L_{SR} = \left[1 + \left(\frac{D_T}{r_0}\right)^{\frac{5}{3}}\right]^{-\frac{6}{5}}$$

Beam wander

Beam wander variance:

$$\langle r_c^2 \rangle = 0.54 \left(H - h_{OGS}\right)^2 \sec^2(\zeta) \left(\frac{\lambda}{2W_0}\right)^2 \left(\frac{2W_0}{r_0}\right)^{5/3}$$

RMS beam wander displacement:

$$\sqrt{\langle r_c^2 \rangle} = 0.69 \left(\frac{\lambda}{2W_0}\right) \left(\frac{2W_0}{r_0}\right)^{5/6}$$

with r_0 the Fried's parameter.

Refractive index structure parameter

HV: Hufnagel-Valley

$$C_n^2(h) = 0.00594 \left(\frac{w}{27}\right)^2 (10^{-5} h)^{10} \exp\left[-\frac{h}{1000}\right] + 2.7 \times 10^{-16} \exp\left[-\frac{h}{1500}\right] + A \exp\left[-\frac{h}{100}\right]$$

with $A = 1.17 \times 10^{-14} \text{m}^{-2/3}$ and $w = 21$ m/s recommended by ITU for night observations.

Rytov variance

$$\sigma_R^2 = 1.23 C_n^2 k^{7/6} L^{11/6}$$

Fried parameter (atmospheric coherence width)

For C_n^2 constant, horizontal propagation with uniform turbulence:

$$r_0 = (0.16 \cdot C_n^2 k^2 L)^{-3/5}$$

For Horizontal link (plane wave):

$$r_0 = \left(1.46 C_n^2 k^2 L\right)^{-3/5}$$

For Vertical link or slant paths:

$$r_0 = \left[0.42 \sec(\zeta) k^2 \int_{h_0}^{H} C_n^2(h) dh\right]^{-3/5}$$

with ζ the zenith angle ($\sec(0) = 1$), h_0 the height above ground level of the uplink transitter and/or downlink receiver, $H = h_0 + L\cos(\zeta)$ is the satellite altitude.

Downlink can be approximated by a plane wave. Use vertical link formula.
Uplink can be approximated by a spherical wave. Use next formula:

$$r_0 = \left[0.42 \sec(\zeta) k^2 \int_{h_0}^{H} C_n^2(h) \left(1 - \frac{h - h_{OGS}}{H - h_{OGS}}\right) dh\right]^{-3/5}$$

with H the satellite altitude, h_{OGS} the altitude of the OGS receiver, and h the integrand.

Greenwood frequency

$$f_g = 2.31 \lambda^{-6/5} \left[\sec(\theta) \int_{h_0}^{H} C_n^2(h) V_{\text{wind}}^{5/3}(h) dh\right]^{3/5}$$

Scintillation

Scintillation index:

$$\sigma_I^2 =$$

Loss:

$$L_{SI} = 4.343\{\ \text{erf}^{-1}(2\rho_{th} - 1) \left[2\ln(\sigma_I^2 + 1)\right]^{\frac{1}{2}} - \frac{1}{2}\ln(\sigma_I^2 + 1)\}$$

Data Communication

Baud Rate

Bandwidth

Modulation

Spectral Efficiency

Capacity

$$C = B \cdot \log_2\left(1 + \frac{S}{N}\right)$$

1 Optical Communication

Signal-to-Noise Ratio

$$\mathrm{SNR}_0 = \frac{i_s}{\sigma_N} = \sqrt{\frac{\eta P_S}{2h\nu B}} \quad \rightarrow \quad 10 \cdot \log(\mathrm{SNR}_0) = 74.9 \text{ dB}$$

$$i_S = \frac{\eta e P_s}{h\nu} = \frac{0.85 \cdot e \cdot 10 \text{ mW}}{h \cdot 193.4 \text{ THz}} = 0.01063$$

$$\left\langle i_N^2 \right\rangle = 2eBi_S = 2 \cdot e \cdot 100 \cdot 10^9 \cdot 0.01063 = 3.406 \cdot 10^{-10}$$

Bit Error Rate

$$\text{4-QAM: } BER \approx \cdot Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

$$\text{16-QAM: } BER \approx 3 \cdot Q\left(\sqrt{\frac{4E_b}{5N_0}}\right)$$

Mean SNR

$$\langle \mathrm{SNR} \rangle = \frac{\mathrm{SNR}_0}{\sqrt{\left(\frac{P_{S0}}{\langle P_S \rangle}\right) + \sigma_I^2(D_G)\mathrm{SNR}_0^2}}$$

$$\langle \mathrm{SNR} \rangle \cong \frac{1}{\sigma_1(D_G)}, \quad \mathrm{SNR}_0 \rightarrow \infty$$

$$\langle \mathrm{SNR} \rangle = \frac{\langle i_S \rangle}{\sigma_N}$$

Fading statistics

$$Pr_{\text{fade}} = 1 - \frac{1}{2}\int_0^{\infty} (u)\text{erfc}\left(\frac{\text{TNR} - \langle\text{SNR}\rangle\, u}{\sqrt{2}}\right) du$$

$$Pr(E) = \langle\text{BER}\rangle = \frac{1}{2}\int_0^{\infty} p_f(u)\text{erfc}\left(\frac{\langle\text{SNR}\rangle\, u}{2\sqrt{2}}\right) du$$

2 Modulation

OOK: On-Off Keying

QPSK: Quadrature Phase Keying

QAM: Quadrature Amplitude Modulation

Data Rates

Photodetectors

Noise

APD

PIN

QPD

Laser transmitter

Amplitude modulation

Phase modulation

Polarization modulation

3 References

?? https://www.nsnam.org/reviews/2016/socis-midterm/fso.html